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Temporal Separation with Entropy Regularization for Knowledge Distillation in Spiking Neural Networks

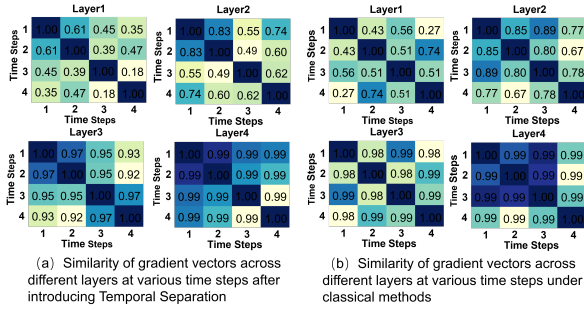


Figure 1. Comparison of gradients between classical methods and methods incorporating the Temporal Separation strategy. The cosine similarity of gradients across different time steps is computed for various layers at different depths using both approaches.

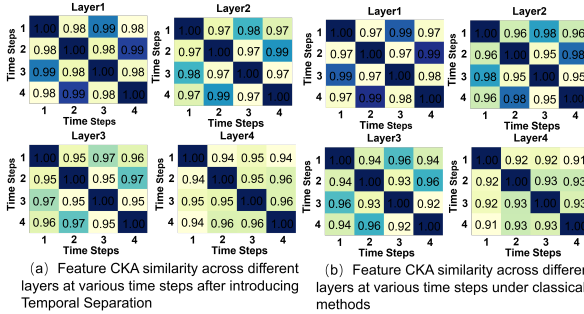


Figure 2. Comparison of similarity of features at different time steps across layers between classical methods and methods incorporating the Temporal Separation strategy.

R1:The Reasons for the Effectiveness of Temporal Separation. As shown in Fig. 1, the introduction of Temporal Separation reduces the gradient similarity between time steps compared to classical methods. This allows each time step to have a distinct optimization direction rather than being broadly optimized towards a potentially incorrect direction. Consequently, this leads to more cohesive and accurate time step features. The strategy introduced in Fig. 2 results in an improvement of 1%-5% in CKA similarity [?], which is consistent with the Fig. 2 in main text. This effectively enhances the accuracy of features across different time steps, thereby improving the overall performance after aggregation.

R1:Design Motivation and Experiments of Entropy Regularization. The introduction of entropy regularization aims to mitigate the impact of high-confidence incorrect predictions from the teacher model. When the value of α is set inappropriately, the learning process of the network may be adversely affected by the erroneous high-confidence

knowledge from the teacher network, leading to optimization bias. Incorporating regularization can alleviate the influence of erroneous information to some extent, thereby enhancing the network performance. Experimental results indicate that adding entropy regularization to the vanilla KD method improves performance by 0.30% to 0.51%.

R1:Firing rate analysis. This was a misstatement. Eq. 12 represents the computation formula for a single layer. The energy consumption analysis in Table. 4 is comprehensive, including all layers except for the final one.

R1:Performance Table is not clear. The term "baseline ANN performance" refers to the accuracy of the teacher network ANN. We aim to compare the learning outcomes of different distillation methods while keeping the teacher network weights fixed. And we will consider your suggestions for future revisions.

R2:Performance Comparison of Transformer-based SNNs. We conducted experiments on the Transformer model using the CIFAR-10/100 datasets. The detailed performance comparisons are presented in Table. 1. Testing results on the ImageNet dataset will be provided in future.

	S-4-256		S-2-384		S-4-384		Time Step
	CF-100	CF-10	CF-100	CF-10	CF-100	CF-10	
Teacher ANN	82.22	96.75	82.22	96.75	82.22	96.75	1
KDSNN	78.38	95.00	79.25	95.59	80.33	95.88	4
LASNN	78.02	94.97	78.91	95.55	79.99	95.79	4
BKDSNN	79.41	95.29	80.63	95.90	81.26	96.06	4
Ours	79.92	95.64	81.07	95.97	81.86	96.72	4

Table 1. Comparison of different distillation methods applied to Transformer-based SNNs on CIFAR10/100 performance.

R2:Inconsistency Between Figures and Equations. We appreciate your careful feedback. We will revise the final paper in accordance with your suggestions.

R3:Analysis of Entropy Regularization. In the main text, Fig. 3 presents a simple experiment on the impact of λ on entropy regularization. A more in-depth theoretical analysis of this aspect will be conducted in future work.

R3:Training Time. In our subsequent experiments, we found that our method is generally faster than vanilla SNN KD training in most cases. This observation contradicts common intuition. I think this discrepancy may be related to the underlying computational logic of PyTorch and CUDA.